EXP 1	Comparative Study of Low-Code Development Versus Traditional Application Development Approaches

Aim:

To compare low-code development and traditional application development approaches based on development speed, ease of use, customization, cost, maintenance, scalability and suitability, using the Hotel Room Booking & Front Office Management application as the reference case.

Theory:

Application development is the process of designing, building, testing and deploying software. Two major approaches are used today.

- **Low-Code Development:** Low-code platforms such as Zoho Creator provide visual builders, drag-and-drop components, pre-built templates and a light scripting language (Deluge). Forms, reports, workflows and dashboards can be built quickly by both professional developers and citizen developers.
- **Traditional Development:** Traditional development uses programming languages such as Java, Python, C# or JavaScript along with frameworks and databases. The developer has full control over the architecture, UI and logic, but needs more time and skill.

Comparison:

Parameter	Low-Code (Zoho Creator)	Traditional Development
Development Time	Days (Hotel app built in hours)	Weeks to months
Coding Requirement	Minimal (Deluge scripts only)	Extensive
Learning Curve	Easy	Moderate to Difficult
Database Design	Visual forms and lookup fields	Manual schema, SQL, ORM
UI Design	Drag-and-drop form/page builder	HTML/CSS/JS frameworks
Customization	Limited to platform features	Highly flexible
Cost	Lower for small projects	Higher initially
Scalability	Moderate	High
Maintenance	Easier, platform managed	Needs skilled developers
Best Suited For	Business apps, internal tools	Complex enterprise systems

Procedure:

1. Study the concepts of low-code and traditional application development.
2. Select a low-code platform (Zoho Creator) and review its features such as form builder, report builder, workflows and page builder.
3. Identify the requirements of the Hotel Room Booking application (rooms, bookings, validations, reports, dashboard).
4. Estimate how the same application would be built with a traditional stack (database, back-end API, front-end UI).
5. Compare both approaches on development time, coding effort, cost, customization, scalability and maintenance.

6. Record the observations and draw conclusions.

Observation:

Hotel App Feature	Low-Code (Zoho Creator)	Traditional
Rooms and Booking forms	Form builder, lookup field	Tables, REST API, HTML forms
Bill calculation	Deluge workflow (few lines)	Back-end code and API
Booking report with filters	Report builder	Custom queries and UI grid
Dashboard with charts	Page builder (Chart, Panel)	Chart library and front-end code
Email confirmation	Workflow email action	SMTP service integration

Result:

The comparative study showed that low-code development (Zoho Creator) enables much faster development of the Hotel Room Booking application with minimal coding, while traditional development offers greater flexibility and scalability for complex systems. Low-code is best suited for business and internal applications.